# An Aperiodic Subwavelength Dipole Structure for Enhancing Near-Field Wireless Power Transfer

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***Abstract*—In this letter, a sparse, aperiodic subwavelength dipole structure (ASDS) is proposed for enhancing wireless power transfer in the near field (NF). The ASDS can have applications in areas such as wireless charging of small electronics, imaging, and proximity sensing. The working principle is inspired by a shifted-beam NF focusing method for similar antenna structures, as well as previous examples of using loaded metawire arrays for far field beamforming. The ASDS is simulated for distances between 1λ and 3λ at 2.4 GHz in the industrial, scientific, and medical band. All dipole elements are parasitically excited by a single source dipole at the center. The design is numerically optimized by adjusting the dipoles' lengths and relative position; a figure of merit is proposed to quantify overall performance. Significant improvement in power transfer efficiency is obtained, compared to reference examples such as a uniformly excited array. The simplicity of the proposed ASDS also makes it advantageous, as no feed networks, multiple excitations, or active components are required.**

***Index Terms*—Aperiodic arrays, arrays, dipoles, focusing, near field (NF), optimization, shifted beams, wireless power transfer (WPT).**

## I. Introduction

IN THIS work, an aperiodic subwavelength dipole structure (ASDS) is proposed for enhancing wireless power transfer (WPT). Using a sparse arrangement of dipoles excited parasitically by a single central element, the radiated fields can be adjusted to maximize power transfer efficiency (PTE). Because the spacing between the dipoles can be quite subwavelength (i.e., $<\lambda/4$) and leads to strong interelement coupling, conventional array theory cannot be used; therefore, the system is studied numerically using the method of moments (MoM). Conceptually, the setup is similar to previous work using impedance-loaded metawire gratings for far-field (FF) beamforming, which can also be adapted for near-field (NF) focusing and WPT by appropriately modifying the optimization objective function [1]. Possible applications include radio frequency identification (RFID), short-to-intermediate range wireless charging of small electronics (e.g., wearable devices, medical implants), imaging, and proximity sensing.

The initial approach of this work is based on the "shifted-beam" (SB) focusing technique, which was developed to achieve subwavelength focusing by using the NF fields of individual dipoles as a basis function to achieve destructive wave interference [2]. This was demonstrated at highly subwavelength distances of $d < \lambda/4$, using arrangements of center-loaded dipoles with spacings of between 1λ and $\lambda/5$ for the purpose of NF imaging. Numeric optimization was used to account for coupling effects as well as the change in input impedance due to changes in the dipole length. Loaded dipole arrays of up to 3 elements along one axis were demonstrated in [3].

The proposed ASDS concept is inspired by the previously discussed concepts and presents a new type of array geometry for WPT enhancement. In this work, it is optimized and studied for NF WPT at distances between 1λ and 3λ. A MoM thin-wire model is implemented in MATLAB to simulate the system. Adjusting the dipoles' position and impedance leads to significant improvements in PTE, superior or comparable to reference cases such as a uniform planar array or a conjugate-phase (CP)-focused array for wavefront matching at the target [4]. The advantage of the ASDS is that it is passive, sparse, and excited by a single input element, forgoing the need for feed networks or multiple input ports. Dipoles are studied in this work primarily for their simplicity, but other antenna designs can also be used.

## II. System Setup

In this section, the system setup studied in this work is discussed. A representative example is shown in Fig. 2. The design frequency is 2.4 GHz, which is in the Industrial, Scientific, and Medical (ISM) radio band and is commonly used for WPT or RFID. A 0.48λ source dipole, together with nonexcited satellite dipoles, constitutes the ASDS. Another 0.48λ dipole is used as the $\mathrm{R_x}$ antenna to determine PTE, by calculating received power at the center where the port is located. Displacements between 1λ and 3λ (5 data points in total, at $\lambda/2$ intervals) between the $\mathrm{T_x}$ and $\mathrm{R_x}$ dipoles are studied.

The input impedance of each dipole can be altered by changing its length as well as its electromagnetic environment by adjusting the dipoles' relative position. Thus, it is possible to optimize for PTE. Coupling between the dipoles can be significant due to their potentially highly subwavelength spacing, meaning that the sinusoidal current approximation, and thus, any analytic solution is not necessarily valid [5]. This is most evident when the dipole lengths deviate significantly from multiples of λ/2. Numeric optimization performed in MATLAB is used to generate the optimal design. Optimization variables used for the

Manuscript received 30 January 2024; accepted 25 February 2024. Date of publication 29 February 2024; date of current version 4 June 2024. *(Corresponding author: Fangwei Chang.)*

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Digital Object Identifier 10.1109/LAWP.2024.3371457

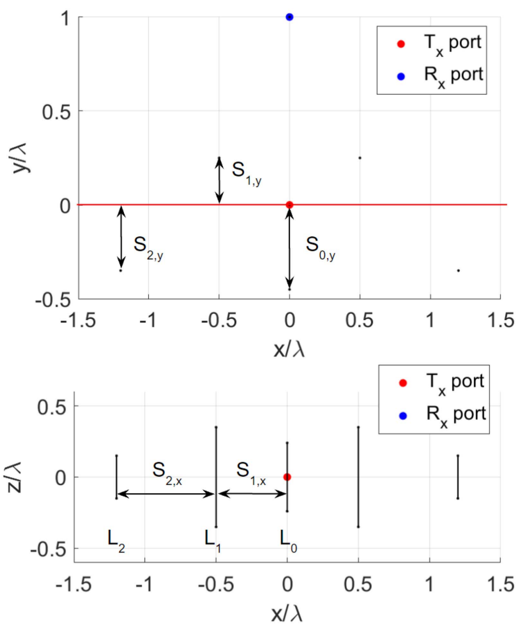


Fig. 1. Representative ASDS structure with labelled optimization variables for clarity. $S_{i,x}$ corresponds to interdipole spacing between two adjacent dipoles; $S_{i,y}$ corresponds to $y$-axis displacement from the plane of the excited $\mathrm{T_x}$ dipole (red line). Variables $L_i$ are the lengths of the dipoles.

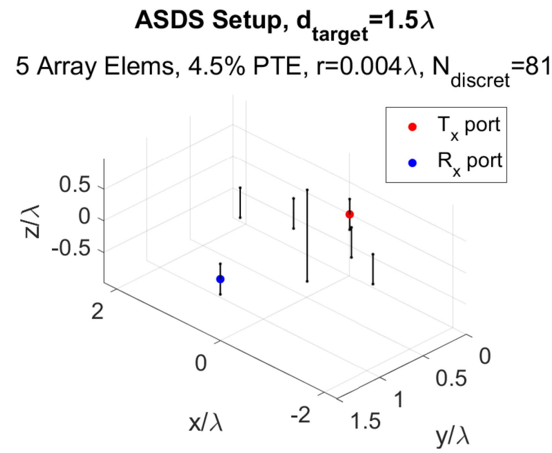


Fig. 2. Optimized 5-element ASDS with $S_x$ and $S_y$ spacing variation for a $\mathrm{R_x}$ element (blue) placed at 1.5λ. The ASDS shown consists of 5 dipoles not including the $\mathrm{T_x}$ dipole (red), with 81 discretizations along each dipole. Parameters for this arrangement are: $L_i = [1.441, 0.473, 0.470]\lambda$, $S_{i,x} = [0.559, 0.734]\lambda$, $S_{i,y} = [0.493, 0.315, 0.500]\lambda$ (labeled in Fig. 1).

system are the dipole lengths $L_1 \ldots L_n$, the interdipole spacings $S_{1,x} \ldots S_{n,x}$ along the $x$-axis, and the dipole shifts $S_{1,y} \ldots S_{n,y}$ along the $y$-axis relative to the source dipole (see Fig. 1).

## III. Optimizing for Power Transfer Efficiency

This section presents the main motive of this work: optimization of the proposed ASDS for enhancing PTE between two dipoles. The system is simulated in MATLAB 2021b using the Hallén integral equation for thin antennas with no loss; a delta-gap voltage excitation at the center is used to excite the source dipole(s) [5]. A dipole radius of $r = 0.004\lambda$ (a typical value used for thin dipoles in simulation [6], resulting in a diameter of ∼1.0 mm at 2.4 GHz) and discretization count of $\mathrm{N_{discret}} = 81$ segments is used to avoid numeric instability, which is significant for $r \geq 0.01\lambda$ as well as for large values of $\mathrm{N_{discret}}$ (i.e., $\mathrm{N_{discret}} > 200$) [6]. For dipole lengths of $L \in [\frac{1}{5}, \frac{3}{2}]\lambda$ considered in this work, this leads to discretizations with lengths in the range of approximately [0.019, 0.002]λ. In general, 60 discretizations are needed for results accurate to within a few percentage points [5].

For simplicity, the centers of all dipole elements in the system are located in the $xy$-plane (i.e., $z = 0$, Fig. 2). Optimization is performed using MATLAB's built-in "fmincon" function, via the gradient descent (GD) method. An optimality tolerance of $10^{-3}$ for the value of the objective function between steps is chosen. This general procedure is also used for all other MATLAB optimizations in this work; the optimization starts with all dipole lengths $L_i$ set to 0.48λ, all $x$-spacings $S_{i,x}$ set to λ/4, and $y$-spacings $S_{i,y}$ set to 0 (i.e., the starting positions of the dipoles are along the $y = 0$ line, Fig. 1). In scenarios where dipoles are spaced very closely (i.e., <λ/20), numeric instability of the calculated currents can occur; appropriately adjusting the allowable range of optimization variables was found to prevent this.

PTE is calculated as the ratio of the received and transmitted (real) power using $\mathrm{P_{real}} = \mathrm{Re(Z)}|\mathrm{I}^2|$, where I is the current measured at the center discretization of the $\mathrm{T_x}$ or $\mathrm{R_x}$ dipole, and Z is the input impedance. The input impedance is determined numerically from $\mathrm{Z = V/I}$ by exciting the corresponding dipole with 1 V at the center, which is verified with full-wave (FW) simulation results using COMSOL Multiphysics 5.6 (RF module). Optimizing for maximum PTE is typically the main goal of WPT systems, although a compromise with maximizing viable transfer distance (application dependent) must also be reached; the two are often mutually exclusive, together with other factors such as the $\mathrm{T_x}$ system's electrical size. The following figure of merit (FOM) is proposed, based on a similar FOM for metamaterial WPT designs [7]:

$$\mathrm{FOM_{WPT}} = \frac{\mathrm{d_{R_x}}}{\mathrm{d_{ASDS}}} \times \mathrm{PTE_{\%}} \tag{1}$$

where $\mathrm{d_{R_x}}$ is the distance between the $\mathrm{T_x}$ and $\mathrm{R_x}$ dipoles and $\mathrm{d_{ASDS}}$ is the length (i.e., greatest transverse dimension) of the ASDS. PTE is a percentage point from 0 to 100. Better performing designs therefore achieve higher PTE over a greater distance, while having a smaller transverse dimension.

### A. Scenarios Studied

Maximizing PTE by adjusting the lengths and spacings of the dipoles in the ASDS is demonstrated. The performance of the ASDS is also compared with reference/benchmark scenarios, such as a uniformly excited array, a CP-focused array, and a single dipole transmitting to another dipole.

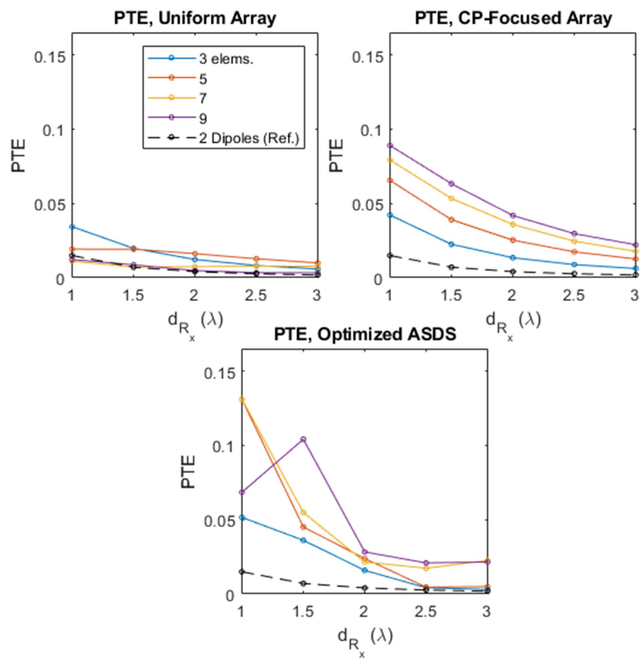


Fig. 3. Comparison of PTE for a uniform array, CP-focused array [4], and optimized ASDS structure for 1–3λ (5 data points). Noticeable improvement is obtained in the case of the ASDS. PTE at each distance point for the ASDS corresponds to the greatest value achieved using optimization in MATLAB.

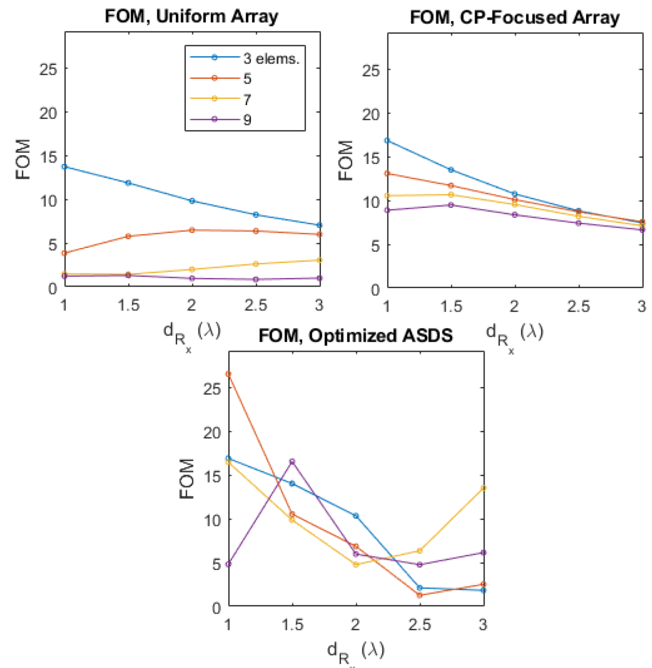


Fig. 4. Comparison of FOM [see (1)] for a uniform array, CP-focused array [4], and optimized ASDS structure for 1–3λ (5 data points). Increased FOM values are observed for the ASDS compared to the uniform array, and performance is comparable to the CP-focused array.

1) *Scenario I. Uniformly excited array:* Consists of center-fed 0.48λ dipoles spaced $\lambda/2$ apart, each excited in-phase with 1 V.
2) *Scenario II. Conjugate-phase (CP)-focused array:* Consists of center-fed 0.48λ dipoles spaced $\lambda/2$ apart, each excited with 1 V. The excitation phases are adjusted to achieve wavefront matching at the $\mathrm{R_x}$ dipole [4].
3) *Scenario III. ASDS:* All elements are allowed to have displacements along the *x*- and *y*-directions. The ASDS has a total of $2\mathrm{N_{side}} + 2$ elements, where $\mathrm{N_{side}}$ is the number of flanking elements on each side not along the *y*–*z* plane (Fig. 2 shows an example with $\mathrm{N_{side}} = 2$). The optimization uses the variables $L_1 ... L_n, S_{1,x} ... S_{n,x}$, and $S_{1,y} ... S_{n,y}$, where the limits are $L_i \in [\frac{1}{5}, \frac{3}{2}]\lambda$, $S_{i,x} \in [\frac{1}{4}, 1]\lambda$, and $S_{i,y} \in [-\frac{1}{2}, \frac{1}{2}]\lambda$.
4) *Scenario IV. Dipole pair:* Consists of a pair of 0.48 λ dipoles, one acting as the $\mathrm{T_x}$ element and the other as the $\mathrm{R_x}$ element (see Fig. 3). Reference case for determining PTE improvement due to addition of ASDS.

Considerable PTE improvement is observed in all studied cases over the reference examples of the uniform array and two-dipole setup (see Figs. 3 and 4). The CP focusing method achieves slightly superior PTE, but in practice, requires a feed network to excite all elements with the appropriate phase and is more difficult to implement; the ASDS has the advantage of being much simpler. Gains in PTE due to the ASDS are greater at shorter distances of $< 1.5\lambda$. Modifying the parameters to test for sensitivity demonstrates that maximum PTE is roughly achieved provided that the variables $(L_i, S_{i,x}, S_{i,y})$ are within ∼10% of their optimized values.

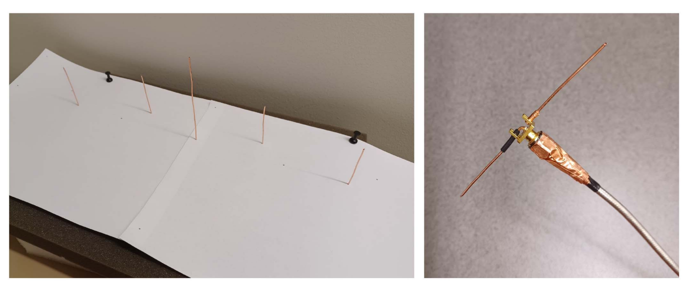

Fig. 5. Experiment setup used to verify ASDS. (Left) ASDS structure, consisting of 18-gauge copper wire segments embedded in polyethylene foam (with printed paper template to accurately specify their locations). (Right) Dipole implementation, constructed using a λ/4 sleeve balun made of copper tape shorted to the outer layer of a 2.9 mm coaxial cable. The dipole arms are cut to length from 18-gauge copper wire and attached to a male 1.6 mm SMA side mount connector, with the lengths adjusted to ensure a resonance frequency of roughly 2.4 GHz.

## IV. Measurement Setup

A selected 5-element ASDS is experimentally validated (see Fig. 2). For simplicity, the $\mathrm{T_x}$ and $\mathrm{R_x}$ 0.48λ dipoles are implemented using $\lambda/4$ sleeve baluns fabricated from a semirigid 2.9 mm coaxial connector cable (see Fig. 5). The theoretical input impedance of the dipoles is ∼73 Ω on their own, which does not necessarily correspond to the dipoles' actual input impedance in the presence of the ASDS or the characteristic

TABLE I
COMPARISON OF PTE CALCULATED/MEASURED BETWEEN MOM CODE, COMSOL FW SIMULATION, AND EXPERIMENT (SETUP FROM FIG. 2)

| | PTE (ASDS, %) | PTE (Ref. Case, %) | Factor of increase of PTE |
|---|---|---|---|
| MoM (lossless) | 4.50 | 0.71 | 6.3 |
| Full-Wave Sim. (incl. loss) | 1.50 | 0.26 | 5.8 |
| Experiment | 1.00 | 0.17 | 5.9 |

TABLE II
RELATIVE ELEMENT CURRENTS FOR OPTIMIZED 9-ELEMENT ASDS STRUCTURES AT SELECT DISTANCES, $r = 10^{-6}\lambda$

| Rel. weight | $d_{\mathrm{R_x}}$ = 1.0λ | $d_{\mathrm{R_x}}$ = 2.0λ | $d_{\mathrm{R_x}}$ = 3.0λ |
|---|---|---|---|
| $\mathrm{w_{center}}$ | 1 | 1 | 1 |
| $\mathrm{w_1}$ | $0.795\angle\mathbf{-197}^\circ$ | $30.790\angle\mathbf{-28}^\circ$ | $16.078\angle\mathbf{-31}^\circ$ |
| $\mathrm{w_2}$ | $1.322\angle\mathbf{-210}^\circ$ | $18.497\angle\mathbf{-208}^\circ$ | $8.680\angle\mathbf{-208}^\circ$ |
| $\mathrm{w_3}$ | $1.698\angle\mathbf{30}^\circ$ | $7.851\angle\mathbf{-29}^\circ$ | $2.936\angle\mathbf{-34}^\circ$ |
| $\mathrm{w_4}$ | $0.904\angle\mathbf{210}^\circ$ | $3.000\angle\mathbf{-210}^\circ$ | $0.628\angle\mathbf{-10}^\circ$ |

50 Ω impedance of the input connector cables. To verify the ASDS straightforwardly, an optimized ASDS example is selected where the input impedance of the $\mathrm{T_x}$ and $\mathrm{R_x}$ dipoles are close to 73 Ω; this corresponds to the setup in Fig. 2, with $\mathrm{Z_{T_x}} = 59.85 + 12.45i\ \Omega$ and $\mathrm{Z_{R_x}} = 82.50 + 19.64i\ \Omega$ as calculated using the MoM code. The mismatch between the dipole and the 50 Ω input line also leads to ≤10% loss theoretically. The unloaded dipoles are made using noninsulated 18-gauge copper wire segments (∼1 mm diameter, i.e., $r \sim 0.004\lambda$) and are secured in a sheet of polyethylene packing foam ($\epsilon_r \sim 1.04$). PTE is measured as $|S_{21}|^2$ using an Agilent E8364B PNA Series Network Analyzer.

For the setup in Fig. 2, the results are compared between the MATLAB MoM simulation (lossless), COMSOL Multiphysics FW simulation (with losses and reflections referenced from the experimental setup), and experiment (see Table I). The proportional improvement in PTE over the reference case of 2 dipoles is specified. The results show reasonable agreement between all three methods when accounting for losses and validates the performance of the ASDS design.

Overall loss of the experimental measurement compared to the MoM result is −6.4 dB, attributable to the nonideal physical setup (i.e., the foam sheet, balun insertion loss, metal losses, reflecting surroundings). Furthermore, each dipole incurs a mismatch loss of approximately −0.7 dB (see Fig. 2). When these factors are accounted for, PTE from the FW simulation increases to 3.7%. The PTE from the MoM simulation is slightly higher due to minor differences in the calculated currents and input impedance at the dipoles' center elements, used to calculate the output/received power as $\mathrm{P_{real}} = \mathrm{Re(Z)}|\mathrm{I}^2|$.

## V. DISCUSSION

The simulated results shown in Figs. 3 and 4 demonstrate considerable improvement in PTE as well as the FOM [see (1)] using the ASDS, compared to the chosen reference examples. Consistent increase of the FOM is observed up to 3λ, beyond which the effect of further optimization or adding elements is not significant. The ASDS uses only a single excited element and is sparse, which is a significant advantage. Although WPT could also be achieved by loading a planar array of dipoles with reactive elements, similar to an embedded electronically steerable passive array radiator (ESPAR), the presented method in this work using length and position adjustment is simpler and achieved PTE is comparable. Furthermore, the ASDS elements reside in the approximate vicinity of the $\mathrm{T_x}$ element, instead of being dispersed along the transmission path. The proportional increase in PTE over the reference case of 2 dipoles (see Table I) is comparable to other WPT designs for a similar distance range [7].

To better understand the behavior of the ASDS, new optimized designs are studied in the limit of an infinitesmally thin dipole ($r = 10^{-6}\lambda$) for select distances (see Table II). The relative current amplitude and relative phase in degrees (compared to the central nonexcited dipole, $L_0$ in Fig. 1) at the center discretization of each dipole is extracted. The results show that all dipoles in this scenario have currents that are either completely in-phase or out of phase, except for dipoles spaced in very close proximity (i.e., <λ/8). This resembles the behavior of the SB method discussed in [2]. Although the WPT distances studied are beyond the distance where beamwidth reduction is significant (i.e., not in the extreme NF), the same weighting scheme nevertheless shows relevance for PTE enhancement.

## VI. CONCLUSION

An ASDS is proposed as a new type of array structure for enhancing WPT. The design is optimized using MATLAB; PTE increase is found to be superior or comparable to reference examples such as the uniform array and the CP-focused array. An advantage associated with the proposed ASDS is that it is simple to implement, with no feed networks or active elements required (whereas the reference examples require all elements to be excited). Experimental verification of a selected 5-element ASDS (see Fig. 2) shows good agreement with MoM and FW simulations, accounting for losses (see Table I).

Possible applications of the ASDS include RFID, wireless charging of small electronics, imaging, or proximity sensing. Dipoles are chosen in this work primarily for their simplicity and compactness, allowing them to be closely spaced, although other antenna types can also be considered. The design can be improved by further increasing the distance of viable WPT, with selective charging for multiple potentially off-center targets. A 2-D or even 3-D layered structure can also be considered, with perpendicularly arranged wires in a grid.

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